

E.G.S. PILLAY ENGINEERING COLLEGE (Autonomous)

NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade Accredited by
NBA T1 Approved by AICTE, New Delhi)



REGULATIONS – R 2023 B.E - BIO MEDICAL ENGINEERING FIRST SEMESTER

COURSE CODE	COURSE NAME	CATEGO RY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
Theory Courses									
2301IP101	Induction Program	-	0	0	0	0	0	0	0
2301MA102	Linear Algebra, Calculus and Partial Differentiation	BSC	3	1	0	4	40	60	100
2301PH101	Physics of Semiconductor and Optoelectronic Devices	BSC	3	0	0	3	40	60	100
2301CH101	Chemistry for Biomedical Engineers	BSC	3	0	0	3	40	60	100
2301GEX04	Problem Solving using Python	ESC	2	0	4	4	50	50	100
2301ENX01	Professional English	HSMC	2	0	0	2	40	60	100
2301TA101	Heritage of Tamils / தமிழர் மரபு	HSMC	1	0	0	1	100	0	100
Laboratory Courses									
2301PHX51	Engineering Physics Laboratory	BSC	0	0	2	1	60	40	100
2301CHX51	Engineering Chemistry Laboratory	BSC	0	0	2	1	60	40	100
2301ENX51	Communication Skills Laboratory	HSMC	0	0	2	1	100	0	100
2301GEX52	Engineering Practices Laboratory	ESC	0	0	4	2	60	40	100
2301LS101	Life Skills – I	-	0	0	0	0	100	0	100
TOTAL			14	1	14	22	720	380	1100

2301MA102	LINEAR ALGEBRA, CALCULUS AND PARTIAL DIFFERENTIATION	L	T	P	C
	(Common to ECE AND BME)	3	1	0	4

PREREQUISITE:

	1. Matrices
	2. Differentiation
	3. Integration.

COURSE OBJECTIVES:

1. To educate Matrix Algebra Technique and curvature theory.
2. To familiarize the students with differential calculus to apply in solving Modern Engineering Problems.
3. To acquaint the student with mathematical tools needed in evaluating multiple integrals and their applications.
4. To familiarize the student with functions of several variables.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Perform matrix diagonalization using principles of eigen values and eigen vectors
CO2:	Develop the evolutes and envelopes of given curves using radius and center of curvature.
CO3:	Apply the multiple integration concepts to solve real life problems involving area and volume
CO4:	Examine the nature of series using comparison, Ratio, Leibnitz tests.
CO5:	Investigate the maxima/minima of multivariable function using partial derivatives and lagrange's method

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			1						1	1
CO2	3	2									1	1
CO3	3	2			1						1	1
CO4	3	2									1	1
CO5	3	2		1	1						1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1		1	
CO2			
CO3		1	
CO4			
CO5		1	

COURSE CONTENTS:

Module I	MATRICES	9 Hours
Rank of a matrix - Consistency of the system of linear equations - linear dependence and independence of vectors. Eigen values and Eigen vectors of a matrix - Cayley-Hamilton theorem and its applications - Reduction to diagonal form - Reduction of a quadratic form to canonical form - orthogonal transformation.		

Module II	DIFFERENTIAL CALCULUS	9 Hours
Curvature in Cartesian co-ordinates – Centre and radius of curvature – Circle of curvature- Evolutes and involutes.		
Module III	MULTIPLE INTEGRALS	9 Hours
Double and triple integrals - computation of surface areas and volumes- Change the order of Integration.		
Module IV	SEQUENCE AND SERIES	9 Hours
Sequences : Definition and Examples – series : Types and Convergence – Series of positive terms – Tests of Convergence: Comparison test, Integral test and D’Alembert’s ratio test – Alternating series – Leibnitz’s test – Series of positive and negative terms – Absolute and conditional convergence.		
Module V	FUNCTIONS OF SEVERAL VARIABLES	9 Hours
Partial differentiation - Total Differentiation – Euler’s theorem and generalization. Maxima and minima of functions of several variables (two and three variables) – Taylor’s series for functions of two variables -Lagrange’s method of Multipliers.		
TOTAL:45 + 15 = 60 Hours		
REFERENCES:		
1.Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi,2018.		
2.R. K. Jain and S. R. K. Iyengar, “Advanced Engineering Mathematics”, 5th ed, Narosa PublishingHouse, 2019.		
3.G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.		
4.Erwin kreyszig, Advanced Engineering Mathematics, 9 th Edition, John Wiley & Sons,2022.		
5.Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11 th Reprint, 2017.		
6.D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2015.		
7.N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.		
8.B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.		

2301PH101	PHYSICS OF SEMICONDUCTOR AND OPTOELECTRONIC DEVICES	L	T	P	C							
	(Common to ECE & BME)	3	0	0	3							
PREREQUISITE:												
	1. Basic knowledge in physics											
COURSE OBJECTIVES:												
1.To instill knowledge on physics of semiconductors, determination of charge carriers and device applications												
2.To make the students to understand the basics of dielectric materials, electrical properties of materials including free electron theory, applications of quantum mechanics												
3.To establish a sound grasp of knowledge on different magnetic & optical properties of materials												
4.To make the students to understand the basics of optoelectronic devices, optical displays and applications												
5.To inculcate an idea of significance of nano structures, quantum confinement and ensuing nano device applications.												
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Understand clearly of semiconductor physics and functioning of semiconductor devices											
CO2:	Apply basics of dielectric materials, gain knowledge on the electrical properties of materials and their applications											
CO3:	Understand the magnetic, optical properties of materials											
CO4:	Demonstrate a strong knowledge in optoelectronic devices and working principles of various optical devices											
CO5:	Apply the importance of nanotechnology and nanodevices											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1								1	1
CO2	3	2	1								1	1
CO3	3	2	1								1	1
CO4	3	2	1								1	1
CO5	3	2	1								1	1
COs Vs PSOs MAPPING												
COURSE CONTENTS:												

Module I	SEMICONDUCTORS AND TRANSPORT PHYSICS	9 Hours
Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap semiconductors – Carrier concentration in intrinsic semiconductors – extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors – Variation of carrier concentration with temperature – Carrier transport in Semiconductors: Drift, mobility and diffusion – Hall effect and devices – Ohmic contacts – Schottky diode.		
Module II	DIELECTRIC MATERIALS AND ELECTRICAL PROPERTIES	9 Hours
Polarization mechanisms: electronic, ionic, orientational, interfacial and total polarization – frequency dependence – local field and Clausius-Mossetti equation – dielectric constant and dielectric loss. Classical free electron theory - Expression for electrical conductivity – Thermal conductivity, expression - Quantum free electron theory - Electron in periodic potential – Energy bands in solids		
Module III	MAGNETIC & OPTICAL PROPERTIES OF MATERIALS	9 Hours
Magnetic materials: Dia, para and ferromagnetic effects – paramagnetism in the conduction electrons in metals – exchange interaction and ferromagnetism – quantum interference devices – GMR devices. Classification of optical materials – Optical processes in semiconductors: optical absorption and emission, charge injection and recombination, optical absorption, loss and gain. Optical processes in quantum wells		
Module IV	OPTOELECTRONIC DEVICES	9 Hours
Optoelectronic devices: light detectors and solar cells – light emitting diode – laser diode – optical processes in organic semiconductor devices –excitonic state – Electro-optics and nonlinear optics: Modulators and switching devices – plasmonics		
Module V	NANO DEVICES	9 Hours
Density of states for solids - Significance between Fermi energy and volume of the material – Quantum confinement – Quantum structures – Density of states for quantum wells, wires and dots – Band gap of nanomaterials –Tunneling – Single electron phenomena – Single electron Transistor- Carbon nanotubes: Properties and applications - Spintronic devices and applications – Optics in quantum structures – quantum well laser.		
TOTAL: 45 Hours		
REFERENCES:		
1.S.O. Kasap. Principles of Electronic Materials and Devices, McGraw Hill Education (Indian Edition), 2020.		
2. R.F.Pierret. Semiconductor Device Fundamentals. Pearson (Indian Edition), 2006.		
3. G.W.Hanson. Fundamentals of Nanoelectronics. Pearson Education (Indian Edition), 2009.		
4. Jasprit Singh, Semiconductor Optoelectronics: Physics and Technology, McGraw- Hill Education (Indian Edition), 2019.		
5. Charles Kittel, Introduction to Solid State Physics, Wiley India Edition, 2019		
6. https://archive.nptel.ac.in/courses/108/108/108108122/		
7. https://onlinecourses.nptel.ac.in/noc20_ph24/preview		

2301CH101	CHEMISTRY FOR BIOMEDICAL ENGINEERS	L	T	P	C							
		3	0	0	3							
PREREQUISITE:												
• Basic knowledge of science up to higher secondary level.												
COURSE OBJECTIVES:												
1. To make the students conversant with boiler feed water requirements, related problems and water treatment techniques.												
2. To impart technological aspects of applied chemistry.												
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Describe the amount of ion present in the water sample											
CO2:	Summarize various polymer materials.											
CO3:	Illustrate biomaterials and corrosion principles.											
CO4:	Describe about drug delivery and photochemistry.											
CO5:	Summarize nano bio technology.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1								1	1
CO2	2	1	1								1	1
CO3	2	1	1								1	1
CO4	2	1	1								1	1
CO5	2	1	1								1	1
COs Vs PSOs MAPPING												
	COs	PSO1	PSO2	PSO3								
	CO1	1										
	CO2	1										
	CO3	1										
	CO4	1										
	CO5	1										
COURSE CONTENTS:												
MODULE I	WATER TECHNOLOGY										9 Hours	
Hardness of water – types – expression of hardness – units – estimation of hardness of water by EDTA -Alkalinity- boiler troubles (scale and sludge) – treatment of boiler feed water – Internal treatment (phosphate, colloidal, sodium aluminate and calgon conditioning) external treatment – Ion exchange process, zeolite process – desalination of brackish water- Reverse Osmosis.												
MODULE II	POLYMER MATERIALS										9 Hours	
Polymer -Classification of polymers – Natural and synthetic; Thermoplastic and Thermosetting Functionality – Degree of polymerization. Properties of polymers: Molecular weight determination weight average - number average. Biopolymers: Structure and biological functions of Collagen, Elastin and Glycoprotein. Synthetic polymers: synthesis properties and biological applications of Polyurethane and polyethylene terephthalate.												

MODULE III	BIOMATERIALS AND CORROSION	9 Hours
Biomaterials: Requirements of Biomaterials, Biological significance of Titanium alloys, Cardio vascular devices, Orthopedic devices. Corrosion: principles of corrosion – PillingBed worth rule – Difference between chemical and electrochemical corrosion – galvanic corrosion – differential aeration corrosion – corrosion control – cathodic protection – sacrificial anodic method.		
MODULE IV	PHOTOCHEMISTRY AND DRUG DELIVERY	9 Hours
Photo Chemistry: Laws of photo chemistry, Electronic Excitation, Jablonski Diagram, Photophysical Processes, Photosensitisation, Photochemical reaction- Chemiluminescence – Introduction to drug Delivery Delivery of drugs: Brain Delivery, Ocular Drug Delivery, Gene Delivery Systems, Vascular Delivery to the Lungs.		
MODULE V	NANO BIOTECHNOLOGY	9 Hours
Nanotechnology: Basics - distinction between molecules, nanoparticles and bulk materials; size-dependent properties. Nano particles: nano cluster, nano rod, nanotube (CNT) and nanowire. Synthetic methods: chemical vapour deposition, laser ablation; synthesis of metal oxide nano particles, conductive nanomaterials, Biological applications of polymer nano composites.		
TOTAL: 45 Hours		
REFERENCES:		
1.Sashi Chawla, A Text book of Engineering Chemistry, Dhanpat Rai Publishing Co., Pvt. Ltd., Educational and Technical Publishers, New Delhi, 3rd Edition, 2015.		
2.S. S. Dara, <i>A Text book of Engineering Chemistry</i> , S. Chand & Co Ltd., New Delhi, 20 th Edition, 2013.		
3.P.C. Jain and Monica Jain, A Textbook of Engineering Chemistry, DhanpatRai publications, New Delhi, 16th edition, 2015.		
4.O.V. Roussak and H.D. Gesser, <i>Applied Chemistry-A Text Book for Engineers and Technologists</i> , Springer Science Business Media, New York, 2nd Edition, 2013.		
5.Alain Nouailhat, “An Introduction to Nanoscience and Nanotechnology”, John Wiley, ISBN:978-1848210073		
6.Nanoparticle Technology for Drug Delivery. Edited by Ram B. Gupta, Uday B. Kompella, 2006, Taylor & Francis Group, 270 Madison Avenue, New York, NY 10016.		
https://onlinecourses.nptel.ac.in/noc23_mm01/preview		
https://onlinecourses.nptel.ac.in/noc23_me46/preview		

2301GEX04	PROBLEM SOLVING USING PYTHON												L	T	P	C
													2	0	4	4
PREREQUISITE:																
The course assumes no prior skill or background in design, art or engineering. It is open to all undergraduates and graduate students with an interest in programming.																
COURSE OBJECTIVES:																
1. To know the basics of problem solving																
2. To learn the basic syntax and semantics of python programming																
3. To acquire programming skills in core python																
4. To use python data structures and develop a skill of designing applications using modules and packages																
COURSE OUTCOME:																
On the successful completion of the course, students will be able to																
CO1:	Identify and apply fundamental problem-solving techniques using algorithms, pseudocode, and flowcharts.															
CO2:	Demonstrate the ability to write, debug, and execute Python programs using basic syntax, control flow, and operators.															
CO3:	Implement Python data structures such as lists, tuples, sets, dictionaries, and arrays for solving real-world problems.															
CO4:	Develop programs using strings, functions, recursion, and lambda expressions to achieve modularity and code reusability.															
CO5:	Apply file handling, exception handling, modules, and packages in Python to design and implement small-scale applications															
COs Vs POs & PSOs MAPPING:																
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	
CO1	3	3	2	2	2								1			
CO2	3	3	2	2	2								1			
CO3	3	3	3	2	2								1			
CO4	3	3	2	2	2								1			
CO5	3	3	3	2	2								1			
COURSE CONTENTS:																
Module I	PROBLEM SOLVING AND PYTHON INTRODUCTION													6 Hours		
Problem Solving: Fundamentals of computing-Algorithms-Building blocks of an algorithm-Pseudocodes and flowcharts. Introduction: Python Interpreter and Interactive mode- Variables and Identifiers- Data Types- Operators-Operator Precedence-Expressions.																
Module II	DECISION MAKING													5 Hours		
Control Flow: If Statement-Elseif Statements-Nested If-else -Loop structure-While Loop-Nested While Loop-For Loop-Nested for Loop- Break and continue statements.																
Module III	DATA STRUCTURES IN PYTHON													7 Hours		
Introduction- Lists: List Operations-List Slicing-List methods- List Loop-Cloning lists- Mutability- Aliasing-Tuples: Tuple Assignment- Tuple as return value- Nested tuples- Basic tuple operations-Advanced list processing- List comprehension -Sets and Dictionaries: Operations and Methods-Arrays.																
Module IV	STRINGS AND FUNCTIONS													6 Hours		
Strings: Introduction, Indexing, Traversing, Concatenating, Appending, Multiplying, Formatting, Slicing, Comparing, Iterating – Basic Built-In String Functions – Functions: Parameters-Return Values-Local and Global Scope-Recursion- Lambda functions.																
Module V	FILES, EXCEPTIONS, MODULES AND PACKAGES													6 Hours		

Files and Exception: Text Files-Reading and writing files-Format operator-command line arguments- errors and exceptions- Handling exceptions – Multiple Exceptions. Modules: Loading and execution-Packages-Python standard Libraries.

LIST OF EXPERIMENTS: **30 Hours**

1. Familiarization with different python IDE
2. Develop simple programs using python syntax and semantics
3. Demonstrate python programs using Arithmetic expressions
4. Illustrate conditional statements with real time problems
5. Basic python applications using list, Tuples.
6. Implement Python program using Dictionaries
7. Implementation of sorting and searching
8. Implement Python program using Strings
9. Write python functions to facilitate code reuse
10. Illustrate file concepts with real time problems
11. Use Exception handling in python applications for error handling
12. Implement simple applications using modules and packages
13. Develop Real Time applications like number guessing, Dice rolling simulator etc.

TOTAL: 60 Hours

REFERENCES:

1. Martin C Brown, “Python The Complete Reference”, Mc Graw-Hill Education – Europe, 4th Edition, 2018
2. Reema Thareja, “Python Programming: Using Problem Solving Approach”, Oxford University Press, 2017.
3. Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, Second Edition, Shroff/O’Reilly Publishers, 2016. (<http://greenteapress.com/wp/thinkpython/>).
4. Ben Stephenson, “The Python workbook A brief introduction with exercises and solutions”, Springer International publishing, Switzerland 2014.
5. Guido van Rossum, Fred L. Drake Jr., “An Introduction to Python – Revised and Updated for Python 3.2”, Network Theory Ltd., 2011.
6. Charles Dierbach, “Introduction to Computer Science using Python”, Wiley India Edition, 2016.
7. Timothy A. Budd, “Exploring Python”, Mc-Graw Hill Education (India) Private Ltd., 2015.
8. <https://nptel.ac.in/courses/106106182>
9. <https://www.learnpython.org/>
10. <https://www.codecademy.com/learn/learn-python>

REQUIREMENTS: (A batch of 30 students)

Hardware Requirements: Standalone Desktop Computer or Server Supporting
Software Requirements: Python Interpreter Version 3

2301ENX01	PROFESSIONAL ENGLISH	L	T	P	C
	Common to B.E /B.Tech Programme (CIVIL,BME,CSE,ECE,EEE,IT,MECH AND AIDS)	2	0	0	2

PREREQUISITE:

	1. Basic English Knowledge
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COURSE OBJECTIVES:

1. To improve the communicative competence of learners.
2. To learn using of basic grammatical structures in suitable contexts.
3. To acquire lexical competence and use them appropriately in a sentence and understand their meaning in a text.
4. To help learners in using the language effectively in professional contexts.
5. To use the language efficiently in expressing their opinions.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to
CO1: Use appropriate words in a professional context
CO2: Gain understanding of basic grammatical structures and use them in right context.
CO3: Read and interpret information presented in tables, charts and other graphic forms
CO4: Write definitions, descriptions, narrations and essays on various topics
CO5: Speak fluently and accurately in formal and informal communicative contexts.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3	1	1
CO2										3	1	1
CO3										3	1	1
CO4										3	1	1
CO5										3	1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-

COURSE CONTENTS:

Module I	FUNDAMENTALS OF COMMUNICATION	9Hours
Reading - Reading brochures (technical context)/ user manuals/, telephone messages / social media messages relevant to technical contexts and emails.		
Writing - Professional emails etiquette, emails / letters (seeking permission for Industrial visit& Complain letter)		
Grammar - Present Tense (simple and progressive); Question types: Why/ Yes or No/ and Tags.		

Vocabulary - Word forms (prefixes& suffixes); Synonyms and Antonyms, Punctuation.		
Module II	NARRATION AND SUMMATION	9 Hours
Reading - Reading longer technical texts (Reading biographies/travelogues/newspaper reports/ travel & technical blogs). Writing - Paragraph writing Short Report on an event (field trip etc.), emails / letters (Writing responses to complaints). Grammar –Past tense (simple); Subject-Verb Agreement. Vocabulary –Preposition, Prepositional Phrases & Phrasal verbs.		
Module III	DESCRIPTION OF A PROCESS / PRODUCT	9 Hours
Reading – Reading advertisements, gadget reviews. Writing – instructions, Checklists, Report Writing (Accident Report & Survey Report (IV)). Grammar –Present & Past Perfect Tenses, Voices (Active ,Passive & Impersonal Passive Voice); Vocabulary –Collocations, Homonyms; and Homophones		
Module IV	CLASSIFICATION AND RECOMMENDATIONS	9 Hours
Reading – Newspaper articles; Journal reports –and Non Verbal Communication (tables, pie charts etc.); Writing- Job / Internship application – Cover letter &Resume, recommendations. Grammar – Articles, Adjectives of Comparison, If conditional sentences Vocabulary –Conjunctions, discourse markers (connectives & sequence words)		
Module V	EXPRESSION	9 Hours
Reading – Company profiles, standard operating procedure (SOP)/ an excerpt of interview with professionals. Writing – Essay Writing (Descriptive or narrative), Grammar – Future Tenses, Numerical adjectives, Relative Clauses. Vocabulary - Cause & Effect Expressions – Content vs Function words.		
TOTAL: 45 Hours		
REFERENCES:		
1. Technical Communication – Principles And Practices By Meenakshi Raman &Sangeeta Sharma, Oxford Univ. Press, 2016, New Delhi.		
2. A Course Book On Technical English By Lakshminarayanan, Scitech Publications (India) Pvt. Ltd.		
3. English For Technical Communication (With CD) By AyshaViswamohan, Mcgraw Hill Education, ISBN : 0070264244.		
4. Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House.		
5. Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi,2003.		
6. Raman. Meenakshi, Sharma. Sangeeta (2019). Professional English. Oxford university press. New Delhi		
7. New Delhi. 2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, NewDelhi.		

2301TA101	தமிழர்மரபு / HERITAGE OF TAMILS							L	T	P	C	
								1	0	0	1	
PREREQUISITE:												
	Cultural and Heritage of Tamil People. And about Sangam Literature, Religion, Art & Architecture during ancient Period. And also about Classical Tamil Literature, Epigraphy, Inscriptions, Non-Tamil source											
COURSE OBJECTIVES:												
1. To preserve and expose the cultural heritage of our ancestors and teachers												
2. To evoke surprise and admiration by creating accessibility to their work, products, and minds.												
3. To promote the practice of science and other cultural activities.												
COURSE OUTCOMES:												
At the end of this course, students will be able to												
CO1:	To use Learning and understanding the development of modern literature.											
CO2:	To acquire the kinds of modern literature and the development of modern literature.											
CO3:	To learn the Tamil Drama Literature, its origin, development, authors Understand the difference between drama of historic period and nowadays. Impact of modern theories in theatre art											
CO4:	To develop the medieval literature and to know the impact of religious literature through the ages											
CO5:	To speak the human values and rights in Tamil literature. Learn to respect the people who live with them Respect the people without the basis Gender, Race and economic status.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3		
CO2										3		
CO3										3		
CO4										3		
CO5										3		
				COs	PSO1	PSO2	PSO3					
				CO1	-	-	-					
				CO2	-	-	-					
				CO3	-	-	-					
				CO4	-	-	-					
				CO5	-	-	-					
COURSE CONTENTS:												
Module I		LANGUAGE AND LITERATURE									3 Hours	
Language Families in India - Dravidian Languages – Tamil as a Classical Language – Classical Literature in Tamil–Secular Nature of Sangam Literature– Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in TamilLand-Bakthi Literature Azhwarsand Nayanmars-Forms of minor Poetry-Development of Modern literature in Tamil-Contribution of Bharathiyar and Bharathidhasan.												

அலகு I மொழி மற்றும் இலக்கியம்: 3 இந்திய மொழிக் குடும்பங்கள் – திராவிட மொழிகள் – தமிழ் ஒரு செம்மொழி – தமிழ் செவ்விலக்கியங்கள் - சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை – சங்க இலக்கியத்தில் பகிர்தல் அறம் – திருக்குறளில் மேலாண்மைக் கருத்துக்கள் – தமிழ்க் காப்பியங்கள், தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம் - பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள் – சிற்றிலக்கியங்கள் – தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி – தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு.		
Module II	HERITAGE-ROCK ART PAINTINGS TO MODERN ART–SCULPTURE	3 Hours
Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridhangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils. அலகு II மரபு – பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை – சிற்பக் கலை: 3 நடுகல் முதல் நவீன சிற்பங்கள் வரை – ஐம்பொன் சிலைகள்– பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் – தேர் செய்யும் கலை – சுடுமண் சிற்பங்கள் – நாட்டுப்புறத் தெய்வங்கள் – குமரிமுனையில் திருவள்ளுவர் சிலை – இசைக் கருவிகள் – மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம் – தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு.		
Module III	FOLK AND MARTIAL ARTS	3 Hours
Therukoothu, Karagattam, Villu Pattu, Kaniyan Koothu, Oyillattam, Leather puppetry, Silambattam, Valari, Tiger dance-Sports and Games of Tamils அலகு III நாட்டுப்புறக் கலைகள் மற்றும் வீர விளையாட்டுகள்: 3 தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான் கூத்து, ஓயிலாட்டம், தோல்பாவைக் கூத்து, சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின் விளையாட்டுகள்.		
Module IV	THINAI CONCEPT OF TAMILS	3 Hours
Flora and Fauna of Tamils & Aham and Puram Concept from Tholkappiyam and Sangam Literature Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age-Export and Import during Sangam Age -Overseas Conquest of Cholas. அலகு IV தமிழர்களின் திணைக் கோட்பாடுகள்: 3 தமிழகத்தின் தாவரங்களும், விலங்குகளும் – தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புறக் கோட்பாடுகள் – தமிழர்கள் போற்றிய அறக்கோட்பாடு – சங்ககாலத்தில் தமிழகத்தில் எழுத்தறிவும், கல்வியும் – சங்ககால நகரங்களும் துறை முகங்களும் – சங்ககாலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி – கடல்கடந்த நாடுகளில் சோழர்களின் வெற்றி.		
Module V	CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE	3 Hours
Contribution of Tamils to Indian Freedom Struggle-The Cultural Influence of Tamils over the other parts of India — Self-Respect Movement – Role of Siddha Medicine in Indigenous Systems of Medicine—Inscriptions & Manuscripts—Print History of Tamil Books.		

அலகு V இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்குத் தமிழர்களின் பங்களிப்பு:	3
இந்திய விடுதலைப்போரில் தமிழர்களின் பங்கு – இந்தியாவின் பிறப்பகுதிகளில் தமிழ்ப் பண்பாட்டின் தாக்கம் – சுயமரியாதை இயக்கம் – இந்திய மருத்துவத்தில், சித்த மருத்துவத்தின் பங்கு – கல்வெட்டுகள், கையெழுத்துப்படிக்கல்கள் - தமிழ்ப் புத்தகங்களின் அச்ச வரலாறு.	
TOTAL:15Hours	
REFERENCES:	
1. தமிழகவரலாறு– மக்களும்பண்பொடும்– மக.மக. பிள்மள (தவளியீடு: தமிழ்நொடுபொடநூல்மற்றும்கல்வியியல்பணிகள்கழகம்).	
2. கணினித்தமிழ்– முமனவர்இல. சுந்தரம். (விகடன்பிரசுரம்).	
3. கீழடி– மவமகநதிக்கமரயில்ெங்ககொலநகரநொகரிகம் (ததொல்லியல்துமற தவளியீடு)	
4. தபொருமந– ஆற்றங்கமரநொகரிகம். (ததொல்லியல்துமறதவளியீடு)	
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)	
6. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies).	
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).	
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies)	
9. Keeladi - 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)	
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Publishedby: The Author)	
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)	
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.	

2301PHX51	ENGINEERING PHYSICS LABORATORY	L	T	P	C
	(Common to all branches)	0	0	2	1

PREREQUISITE:

	Basic knowledge in physics
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COURSE OBJECTIVES:

1. To learn the proper use of various kinds of physics laboratory equipment
2. To learn how data can be collected, presented and interpreted in a clear and concise manner.
3. To learn problem solving skills related to physics principles and interpretation of experimental data.
4. To determine error in experimental measurements and techniques used to minimize such error.
5. To make the student an active participant in each part of all lab exercises

COURSE OUTCOMES:

On the successful completion of the course, students will be able to	
CO1:	Apply the theoretical concepts of physics in procedures and techniques in performing the experiments
CO2:	Apply and demonstrate, thermal conductivity, electrical properties of metals and semiconductors, elastic properties of materials and oscillations through experiential learning
CO3:	Demonstrate the use of monochromatic light, lasers in optical fiber communication and quantum mechanics towards specific engineering applications
CO4:	Use different measuring devices/ meters to record the data with precision and apply the mathematical concepts/equations to obtain quantitative results
CO5:	Develop basic communication skills through working in groups in performing the laboratory experiments and by interpreting the results

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	1	1	2	2			
CO2	3	3	2	2	1	1	1	2	2			
CO3	3	3	2	2	1	1	1	2	2			
CO4	3	3	2	2	1	1	1	2	2			
CO5	3	3	2	2	1	1	1	2	2			

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	2		
CO2	2		
CO3	2		
CO4	2		
CO5	2		

LIST OF EXPERIMENTS (Any 8 experiments to be performed)

1. Torsional pendulum - Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.
2. Simple harmonic oscillations of cantilever.

3. Non-uniform bending - Determination of Young's modulus
4. Uniform bending – Determination of Young's modulus
5. Laser- Determination of the wavelength of the laser using grating
6. Air wedge - Determination of thickness of a thin sheet/wire
7. a) Optical fibre -Determination of Numerical Aperture and acceptance angle b) Compact disc- Determination of width of the groove using laser.
8. Acoustic grating- Determination of velocity of ultrasonic waves in liquids.
9. Ultrasonic interferometer – determination of the velocity of sound and compressibility of liquids
10. Determination of Band gap of a semiconductor.
11. Poiseuille's method for finding viscosity of a liquid
12. Lee's Disc-Thermal conductivity of bad conductor
13. Spectrometer-determination of wavelength using grating
Total :30 Hours

References
1. Practical Physics', R.K. Shukla, AnchalSrivastava, New age international (2011)
2. B.Sc. Practical Physics', C.L Arora, S. Chand &Co. (2012)

2301CHX51	ENGINEERING CHEMISTRY LABORATORY	L	T	P	C
	COMMON TO ALL BRANCHES	0	0	2	1

PREREQUISITE:

	Basic knowledge of science up to higher secondary level
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COURSE OBJECTIVES:

1.	To make the students conversant with boiler feed water requirements, related problems and water treatment techniques
2.	To impart technological aspects of applied chemistry

COURSE OUTCOMES:

On the successful completion of the course, students will be able to	
CO1:	Estimate the amount of ion present in the water sample.
CO2:	Determine the pH of the solutions.
CO3:	Estimate the corrosion behavior of metals.
CO4:	Determine the acid content using electrochemical principles.
CO5:	Determine the molecular weight of the polymer.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			1			1	1			
CO2	3	2			1			1	1			
CO3	3	2			1			1	1			
CO4	3	2			1			1	1			
CO5	3	2			1			1	1			

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1		
CO2	1		
CO3	1		
CO4	1		
CO5	1		

LIST OF EXPERIMENTS

1.	Determination of total, temporary & permanent hardness of water by EDTA method
2.	Comparison of alkalinities of the given water samples
3.	Estimation of iron content of the given solution using potentiometer
4.	Corrosion experiment – weight loss method
5.	Conductometric titration of strong acid Vs strong Base
6.	Determination of molecular weight of a polymer by viscometry method

7. Determination of percentage of copper in alloy
8. Determination of ferrous iron by Spectrophotometry method
9. Estimation of calcium present in cement.
10. Determination of strength of given hydrochloric acid using pH meter
11. Estimation of sodium ion present in water by flame photometer.
12. Estimation of dissolved oxygen in a water sample/sewage by Winklers method.
13. Synthesis of metal oxide nanoparticles by chemical method.
Total :30 Hours

REFERENCES:
1. Experimental organic chemistry, Daniel R. Palleros, John Wiley & Sons, Inc., New York (2001)
2. Engineering Chemistry”, Jain & Jain, 15th edition, Dhanpat Rai Publishing company, New Delhi
3. Vogel’s Textbook of practical organic chemistry, Furniss B.S. Hannaford A.J, Smith P.W.G and Tatchel A.R LBS Singapore (1994)
4. LBS Singapore (1994). Kolthoff I.M., Sandell E.B. et al Mcmillan, Madras 1980

2301ENX51	COMMUNICATION SKILLS LABORATORY	L	T	P	C
	Common to B.E /B.Tech Programme (CIVIL,BME,CSE,ECE,EEE,IT,MECH and AI&DS)	0	0	2	1

PREREQUISITE:

	Basic English Knowledge
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COURSE OBJECTIVES:

1. To facilitate computer-aided multi-media instruction enabling individualized and independent language learning
2. To bring about a consistent accent and intelligibility in their pronunciation of English by providing an opportunity for practice in speaking.
3. To prepare them to use communicative language and participate in different types of speaking environments.
4. To expose the Students to participate in group discussions, debates with ease.
5. To enable the students become strong in LSRW skills.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to	
CO1:	Improve their listening, reading, speaking and writing skills.
CO2:	Develop their communication competency.
CO3:	Use language effectively in professional contexts.
CO4:	Develop the ability to face campus interviews.
CO5:	Use language efficiently in expressing their opinions

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3	1	1
CO2										3	1	1
CO3										3	1	1
CO4										3	1	1
CO5										3	1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-

COURSE CONTENTS:

Module I	LISTENING	6 Hours
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Listening for general information-specific details Audio / video (formal & informal). Listening IELTS/TOFEL/ TED Talks and educational videos. Listening to podcasts, anecdotes / stories / event narration; documentaries and interviews with celebrities. Listening - Listen to product and process descriptions; and advertisements about products. Listening – Listening to debates/ discussions; different viewpoints on an issue; and panel discussions.		
Module II	SPEAKING	6 Hours
Self-Introduction - Role play exercises based on workplace contexts- Group discussion (Discussing advantages and disadvantages/ purposes and reasons)- Discussing progress toward goals- Discussing past events in life-Making telephone calls (politeness strategies- making polite requests, making polite offers, replying to polite requests and offers) Interpreting (Picture, locations in workplaces).		
Module III	READING	6 Hours
Reading– Intensive Reading -Comprehending general and technical articles -Cloze reading - Inductive reading- Short narrative and descriptions from newspapers – Skimming and scanning- reading and interpretation-Critical reading Interpreting and transferring graphical information- Sequencing of sentences..		
Module IV	WRITING	6 Hours
Writing- Precise writing –Summarizing- Interpreting visual texts (pie chart, bar chart, picture, advertisements etc., - Proposal writing (launching new units or department in a institution or industry & to get loan from bank) -Report writing (accident, progress, project, survey, Industrial visit)- Job application-Resume.		
Module V	PERSONALITY DEVELOPMENT	6 Hours
Introduction to life skills -emotional intelligence (visualizing and experiencing purpose)-Self-awareness - Time management-Stress management -Leadership- teamwork & dealing with ambiguity--interview planning- Mock Interviews-- Self-Concept. Organizational etiquette.		
TOTAL: 30 HOURS		
REFERENCES:		
1.Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House.		
2.Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi,2003.		
3.New Delhi. 2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, NewDelhi.		
4.Developing Communication Skills by Krishna Mohan, MeeraBannerji- Macmillan India Ltd. 1990, Delhi.		
5.Business Correspondence and Report Writing by Prof. R.C. Sharma & Krishna Mohan, Tata McGraw Hill & Co. Ltd., 2001, New Delhi.		
https://swayam.gov.in/explorer?searchText=english (Link for NPTEL/SWAYAM/MOOC Courses)		
https://ieltsionlinetests.com (Link for modern tool usage)		

2301GEX52		ENGINEERING PRACTICES LABORATORY											L	T	P	C	
														0	0	4	2
PREREQUISITE: NIL																	
COURSE OBJECTIVES:																	
1. To provide hands on training for fabrication of components using sheet metal and welding equipment / tools.																	
2. To develop skill for using carpentry and fitting tools to make simple components and metal joints.																	
3. To provide training for making simple house hold pipe line connections using suitable tools.																	
4. To develop the skill to make / operate/utilize the simple engineering components.																	
COURSE OUTCOMES:																	
At the end of this course, Students will be able to,																	
CO1: Fabricate simple components using sheet metal using suitable tools.																	
CO2: Prepare simple components using suitable fitting tools.																	
CO3: Fabricate simple components using welding equipments.																	
CO4: Make simple components / joints using carpentry power tools.																	
CO5: Make simple house hold pipe line connections using suitable tools.																	
COs Vs POs & PSOs MAPPING:																	
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3		
CO1	2								2			1	-	-	-		
CO2	2	1							2			1	-	-	-		
CO3	2	1			1				2			1	-	-	-		
CO4	2	1			1				2			1	-	-	-		
CO5	2								2			1	-	-	-		
LIST OF EXPERIMENTS																	
1. Forming of simple object in sheet metal using suitable tools.(Example: Dust Pan, Rectangular tray and Cone making)														6 Hours			
2. Prepare V (or) Half round (or) Square (or) Dovetail joint from the given mild Steel flat.														5 Hours			
3. Fabrication of a simple component using thin and thick plates using arc welding. (Example: Butt , Lap and T - Joints)														6 Hours			
4. Making a simple component using carpentry power tools.(Example: Cross Lap, T-Lap, Dove tail joints and Electrical switch box / Tool box / Letter box)														6 Hours			
5. Construct a household pipe line connections using pipes, Tee joint, four way joint, elbow, union, bend, Gate valve and Taps.														5 Hours			
6. Study of gas welding equipment and its demonstration.														2 Hours			
Total : 30 Hours																	
References:																	
1. S. Gowri &T.Jeyapoovan, “Engineering Practices Lab Manual” 5th Edition,Vikas Publishing.																	
2. Dr. V. Ramesh Babu,”Engineering Practices Laboratory Manual” Revised Edition 2019-20, VRB Publishers Pvt. Ltd.																	